

Fundamental Problems

COMPCSI 752: Big Data Management

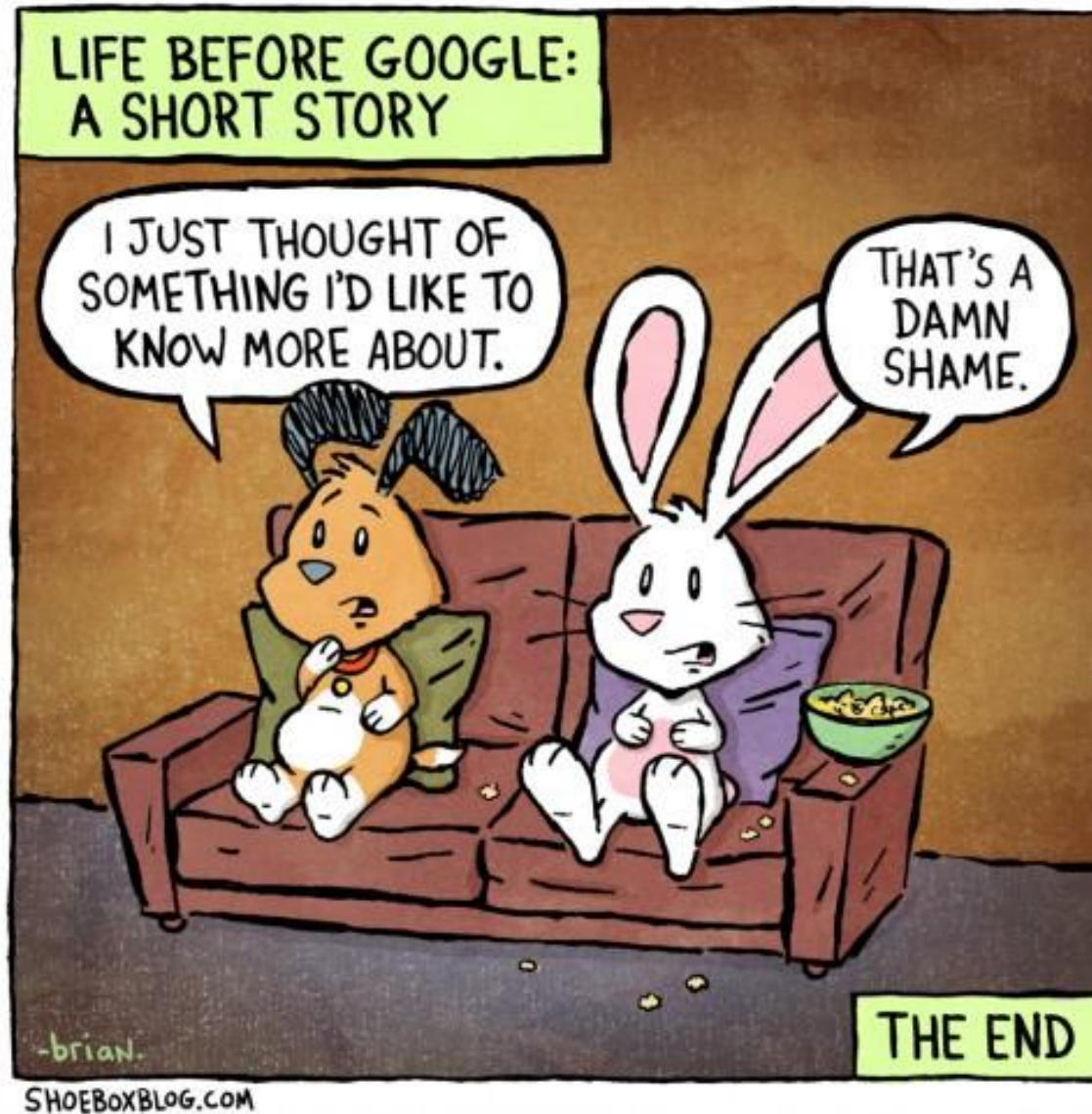
University of Auckland

Credit to Ninh Pham for these slides

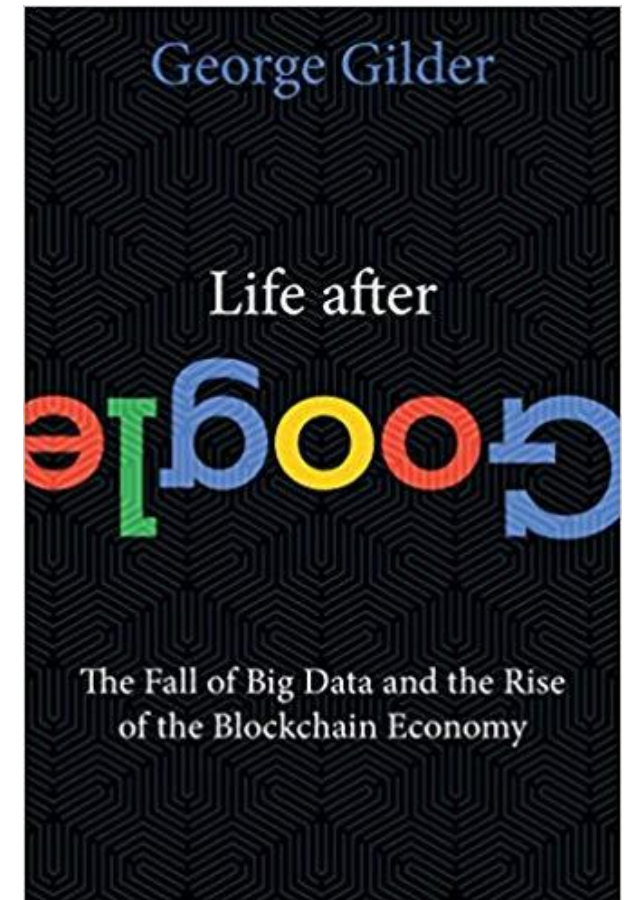
Outline

- Web crawling
 - Near-duplicate detection
- Web search
 - PageRank
- Data cleaning
 - tf-idf representation
 - Similarity join

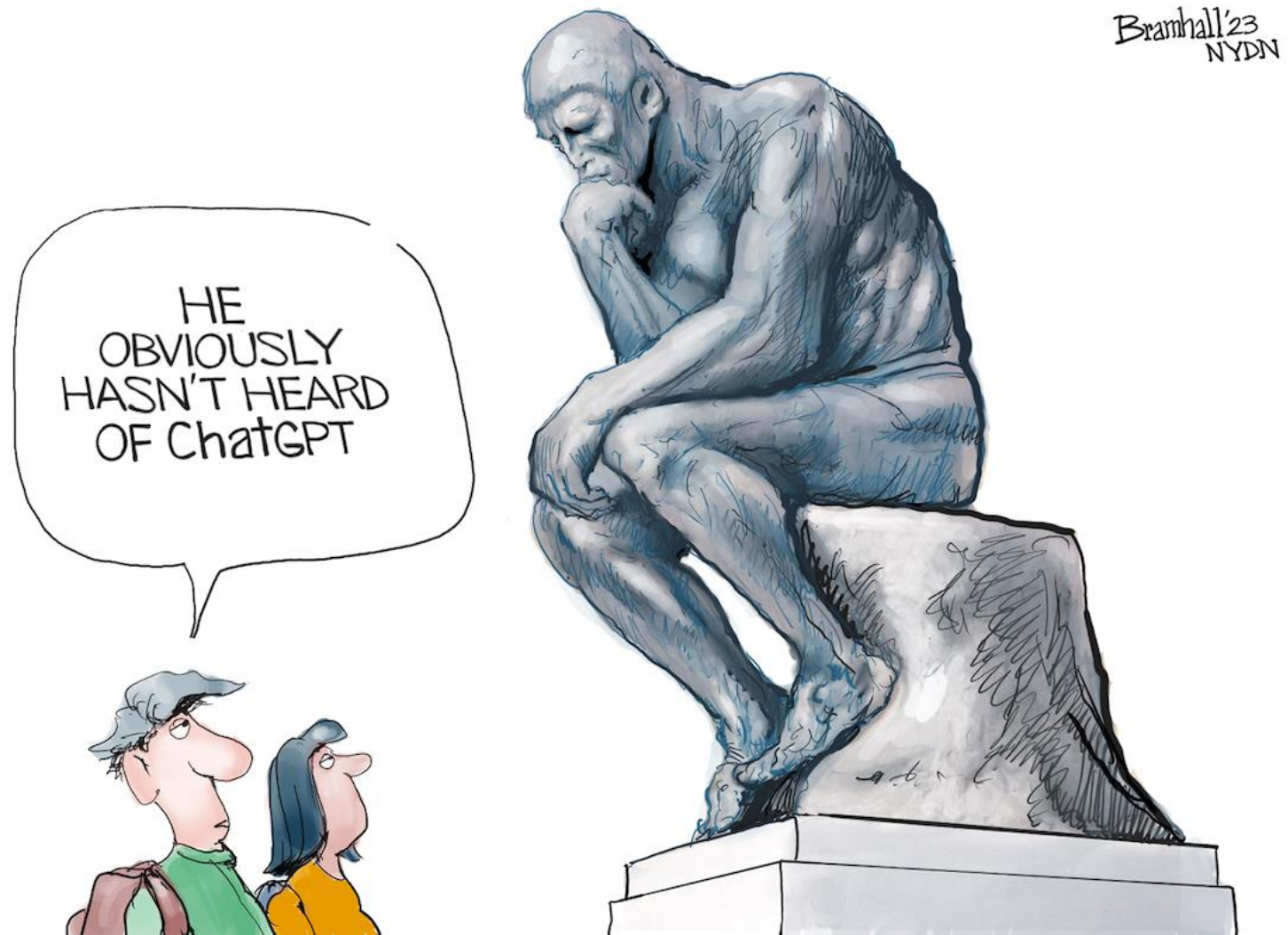
Life before Google



Life after Google



Life after ChatGPT



<https://www.dallasnews.com/opinion/commentary/2023/02/23/editorial-cartoon-chatgpt/>



"university of Auckland"



All

Maps

Images

News

Videos

More

Settings

Tools

About 9,460,000 results (0.69 seconds)

The University of Auckland | NZ's Leading University | auckland.ac.nz

[Ad] www.auckland.ac.nz/ ▼

Ranked #1 in NZ for graduate employability. Top 100 World QS Rankings. #1 Graduate Employability. Highest Ranked NZ Uni. 4 Main Campuses in Akld. Degree programs: Undergraduate, Postgraduate, Masters, PhDs.

Open Day

Get a taste of your future and explore our campus at our Open Day.

PG Week

Join us at PG Week and be inspired to study at NZ's leading university

The University of Auckland: Welcome

<https://www.auckland.ac.nz/en.html> ▼

The University of Auckland is New Zealand's world-ranked university.

Study options

See our undergraduate and postgraduate study options ...

Contact us

Contact us. Have an enquiry? Reach out to our help centres ...

Postgraduate study options

Postgraduate prospectuses - Masters - Doctoral programmes

[More results from auckland.ac.nz »](#)

Apply now

Application for Admission - How to apply - Scholarships and awards

Undergraduate study options

Conjoint programmes - General Education - Open Day - ...

International students

Study options - Entry requirements - How to apply - ...

University of Auckland - Wikipedia

https://en.wikipedia.org/wiki/University_of_Auckland ▼

The **University of Auckland** is the largest university in New Zealand, located in the country's largest city, Auckland. It is the highest-ranked university in the ...

Students: 33,050 (EFTS, 2015) Vice-Chancellor: **Stuart McCutcheon**

Chancellor: Scott St John Motto in English: By natural ability and hard work



The University of Auckland

Website

Directions

Save

University

The University of Auckland is the largest university in New Zealand, located in the country's largest city, Auckland. It is the highest-ranked university in the country, being ranked 85th worldwide in the 2018/19 QS World University Rankings. [Wikipedia](#)

Address: Auckland, 1010

Total enrollment: 41,953 (2014)

Undergraduate tuition and fees: Domestic tuition: 6,000 NZD (2018), International tuition: 33,688 NZD (2018)

[Suggest an edit](#)

Events

Sun, 24 Mar 10:30 pm UoA Greens on Campus AGM 2019

Wed, 27 Mar 6:30 pm Simon Ewings Public Lecture

Fri, 29 Mar 4:30 pm Metrogaine



"university of Auckland"



All

Maps

Images

News

Videos

More

Settings

Tools

About 9,460,000 results (0.69 seconds)

~~The University of Auckland | NZ's Leading University | auckland.ac.nz~~

(Ad) www.auckland.ac.nz/ ▼

Ranked #1 in NZ for graduate employability. Top 100 World QS Rankings. #1 Graduate Employability. Highest Ranked NZ Uni. 4 Main Campuses in Akld. Degree programs: Undergraduate, Postgraduate, Masters, PhDs.

Open Day

Get a taste of your future and explore our campus at our Open Day.

PG Week

Join us at PG Week and be inspired to study at NZ's leading university

The University of Auckland: Welcome

<https://www.auckland.ac.nz/en.html> ▼

The University of Auckland is New Zealand's world-ranked university.

Study options

See our undergraduate and postgraduate study options ...

Contact us

Contact us. Have an enquiry? Reach out to our help centres ...

Postgraduate study options

Postgraduate prospectuses - Masters - Doctoral programmes

[More results from auckland.ac.nz »](#)

Apply now

Application for Admission - How to apply - Scholarships and awards

Undergraduate study options

Conjoint programmes - General Education - Open Day - ...

International students

Study options - Entry requirements - How to apply - ...

University of Auckland - Wikipedia

https://en.wikipedia.org/wiki/University_of_Auckland ▼

The **University of Auckland** is the largest university in New Zealand, located in the country's largest city, Auckland. It is the highest-ranked university in the ...

Students: 33,050 (EFTS, 2015) Vice-Chancellor: [Stuart McCutcheon](#)

Chancellor: Scott St John Motto in English: By natural ability and hard work



The University of Auckland

Website

Directions

Save

University

The University of Auckland is the largest university in New Zealand, located in the country's largest city, Auckland. It is the highest-ranked university in the country, being ranked 85th worldwide in the 2018/19 QS World University Rankings. [Wikipedia](#)

Address: Auckland, 1010

Total enrollment: 41,953 (2014)

Undergraduate tuition and fees: Domestic tuition: 6,000 NZD (2018), International tuition: 33,688 NZD (2018)

[Suggest an edit](#)

Events

Sun, 24 Mar 10:30 pm UoA Greens on Campus AGM 2019

Wed, 27 Mar 6:30 pm Simon Ewings Public Lecture

Fri, 29 Mar 4:30 pm Metrogaine

Page 10 of about 95 results (0.77 seconds)

The University of Auckland | Start your journey to study | auckland.ac.nz

Ad www.auckland.ac.nz/ ▼

Open doors with a world-class degree from NZ's leading university. Apply Now! 4 Main Campuses in Akld. #1 Graduate Employability. Top 100 World QS Rankings. Highest Ranked NZ Uni. Degree programs: Undergraduate, Postgraduate, Masters, PhDs.

Did judgmental gods help societies grow? | Science | AAAS

<https://www.sciencemag.org/news/2019/03/did-judgmental-gods-help-societies-grow> ▼

3 days ago - ... adds Quentin Atkinson, an evolutionary psychologist at the University of Auckland in New Zealand who wasn't involved in the new research.

Gbi staff directory - African Union Youth Division

<https://youth.africa-union.org/eqb6aih/ghmrhq.php?gyrydtzea=gbi-staff-directory> ▼

... The Global Business and Innovation (GBI) program is a collaboration with the Chinese University of Hong Kong – Shenzhen and the University of Auckland.

Intellect Books | Choreographic Practices

<https://www.intellectbooks.com/choreographic-practices> ▼

Carol Brown University of Auckland. Franc Chamberlain University of Huddersfield. Emilyn Claid University College Falmouth. Darcey Callison York University.

E cigarettes auckland - kameographics

kameographics.com/dv6fo2t/cj5rryw.php?gyrydtzea=e-cigarettes-auckland ▼

Items 1 - 51 of 193 - Participants sought for e-cigarette stop-smoking trial 19 April 2017 A clinical trial by the University of Auckland is calling for smokers to take ...

University of birmingham ranking qs

atk.egemdekorasyon.com/zxyz1wd/fcnuzst.php?gyrydtzea=university-of...ranking... ▼

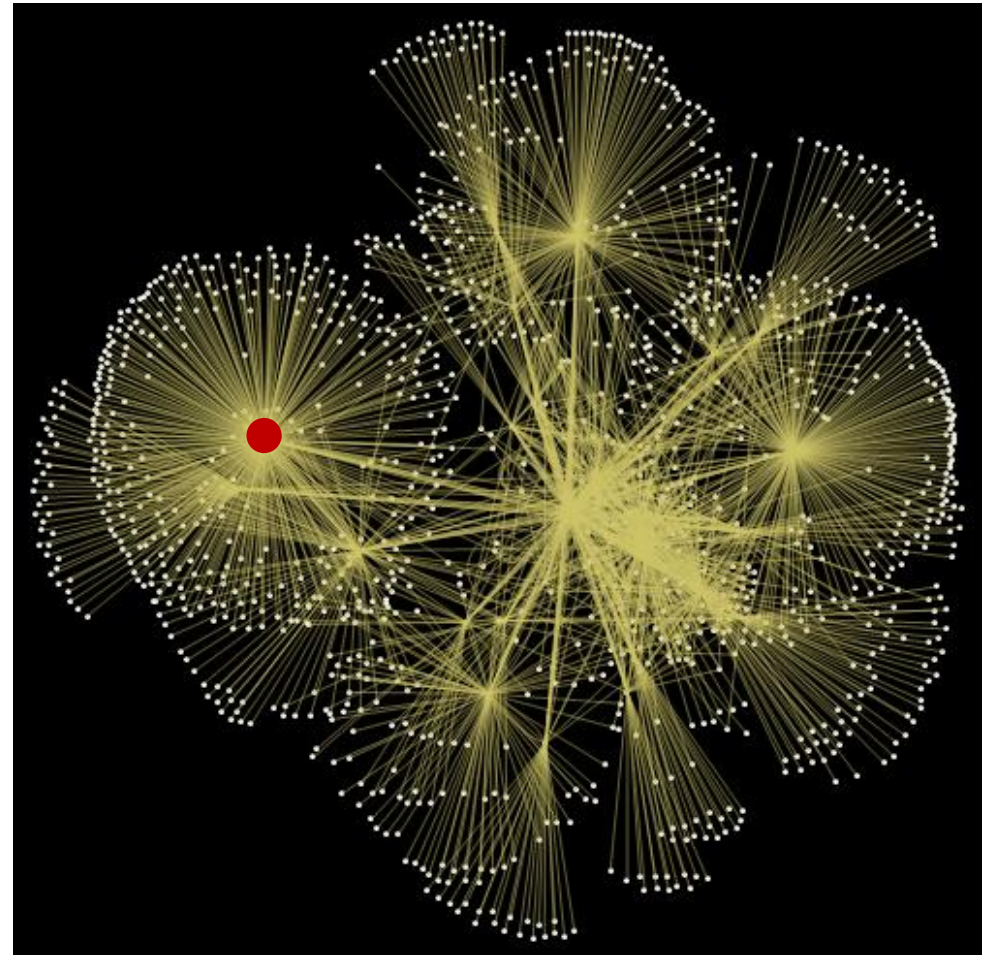
The University of Auckland is New Zealand's leading university in the QS World University Rankings 2019. com is one of the top international rankings ...

Use structure of web to rank pages

- Using the graph structure of the web to assign some score to a page and rank the page by its score

Use structure of web to rank pages

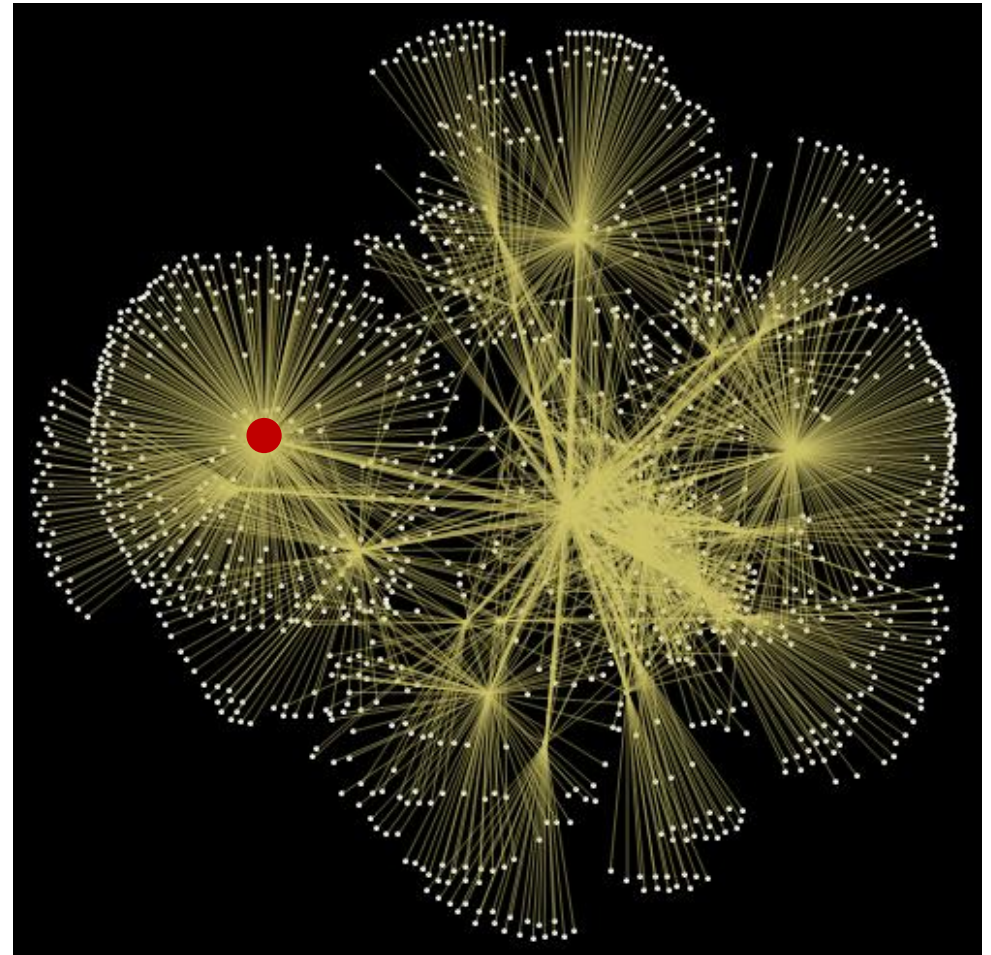
- Using the graph structure of the web to assign some score to a page and rank the page by its score
 - Vertices: webpages
 - Edges: hyperlinks



<https://www.wolfram.com/mathematica/new-in-8/graph-and-network-modeling/structure-of-the-web.html>

Use structure of web to rank pages

- Using the graph structure of the web to assign some score to a page and rank the page by its score
 - Vertices: webpages
 - Edges: hyperlinks
- General ideas:
 - If a page is linked by a large number of important pages, it is itself important

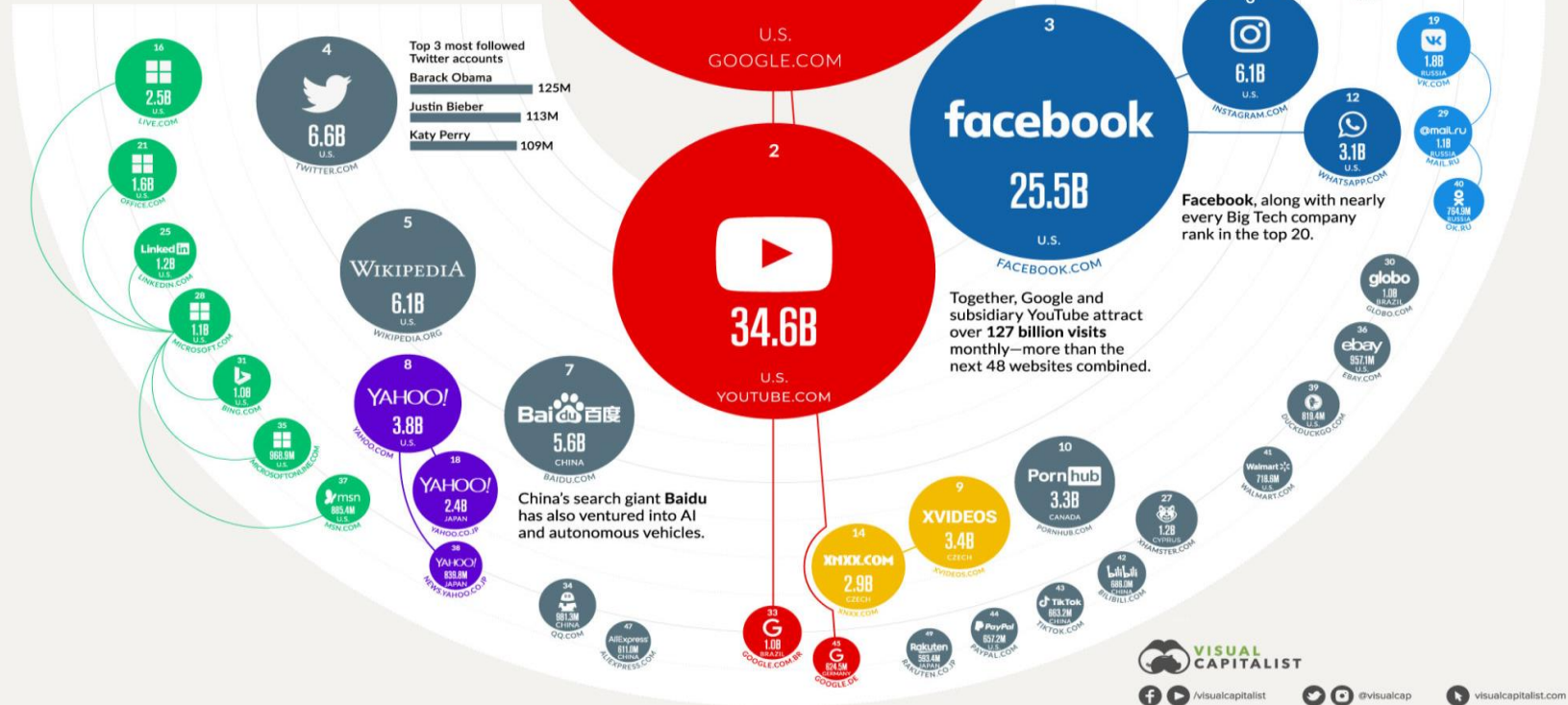
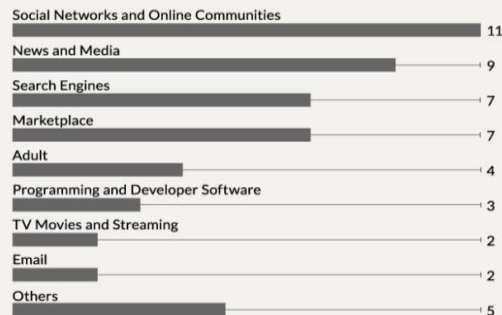


<https://www.wolfram.com/mathematica/new-in-8/graph-and-network-modeling/structure-of-the-web.html>

THE WORLD'S Top 50 Websites

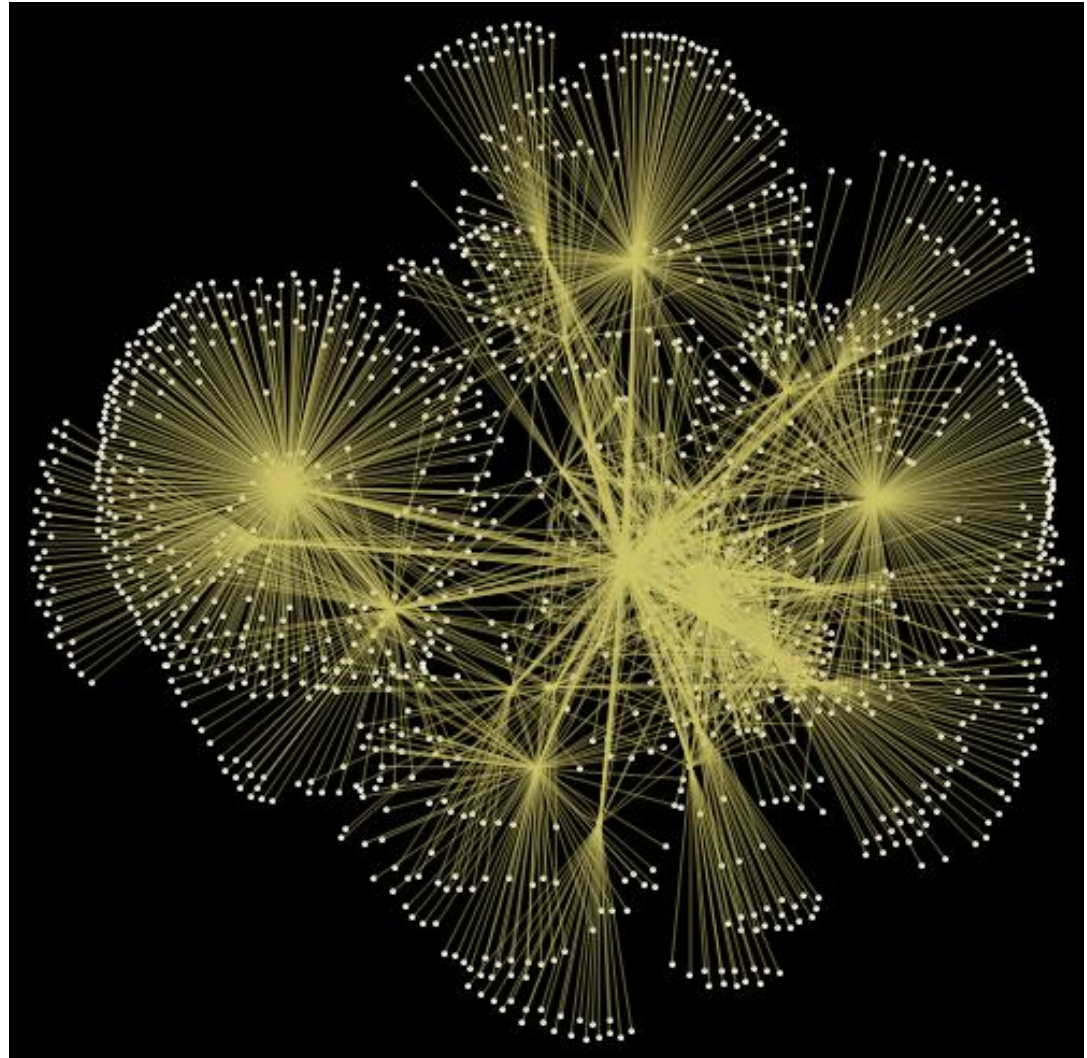
Below, we show the key players—from Google to Twitter—that currently dominate the Internet.

BREAKDOWN BY CATEGORIES (GLOBAL, NOV 2020)



From general idea to Page Rank

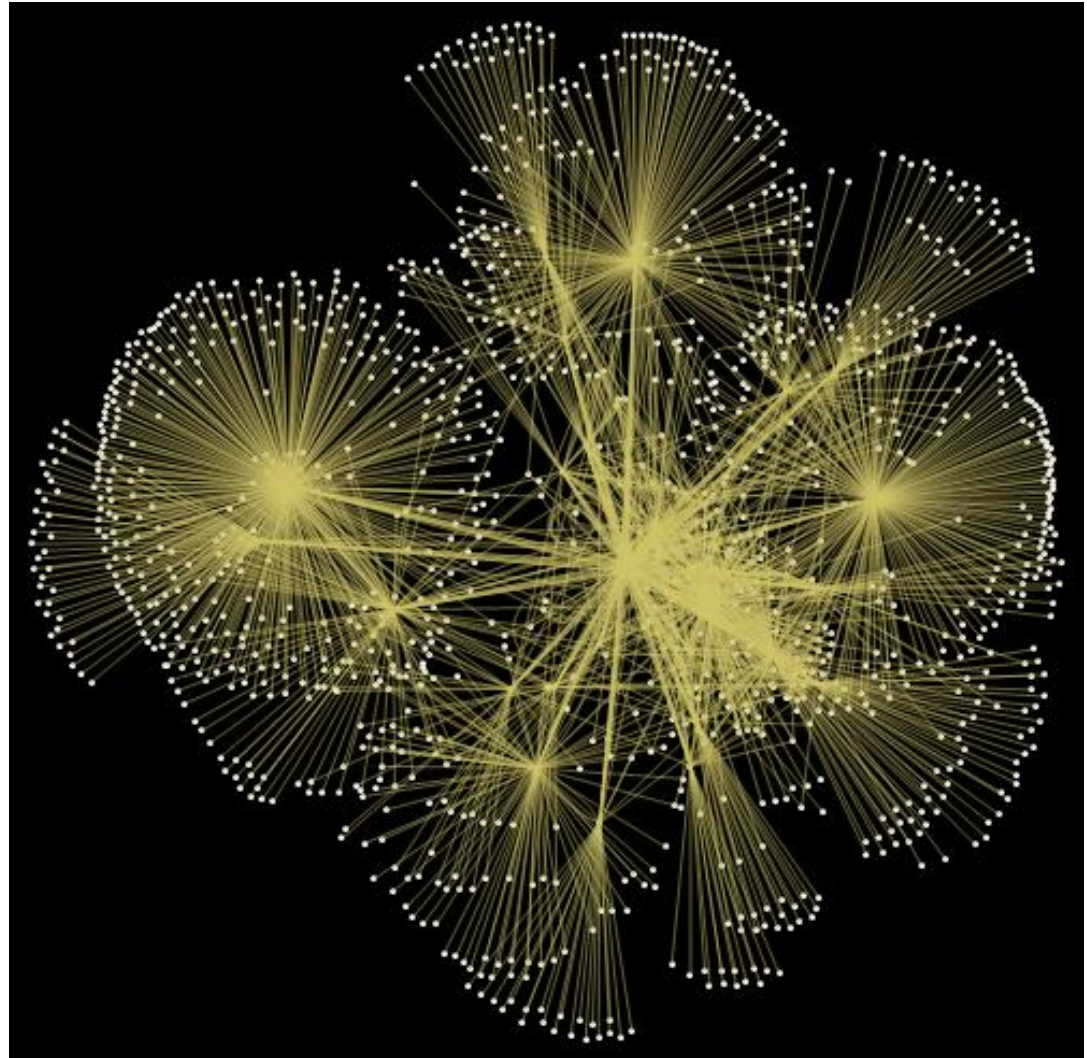
- Why not using the number of in-links as a score?



<https://www.wolfram.com/mathematica/new-in-8/graph-and-network-modeling/structure-of-the-web.html>

From general idea to Page Rank

- Why **not** using the number of in-links as a score?
 - No, “spam farm” ☹️



PageRank on directed graphs

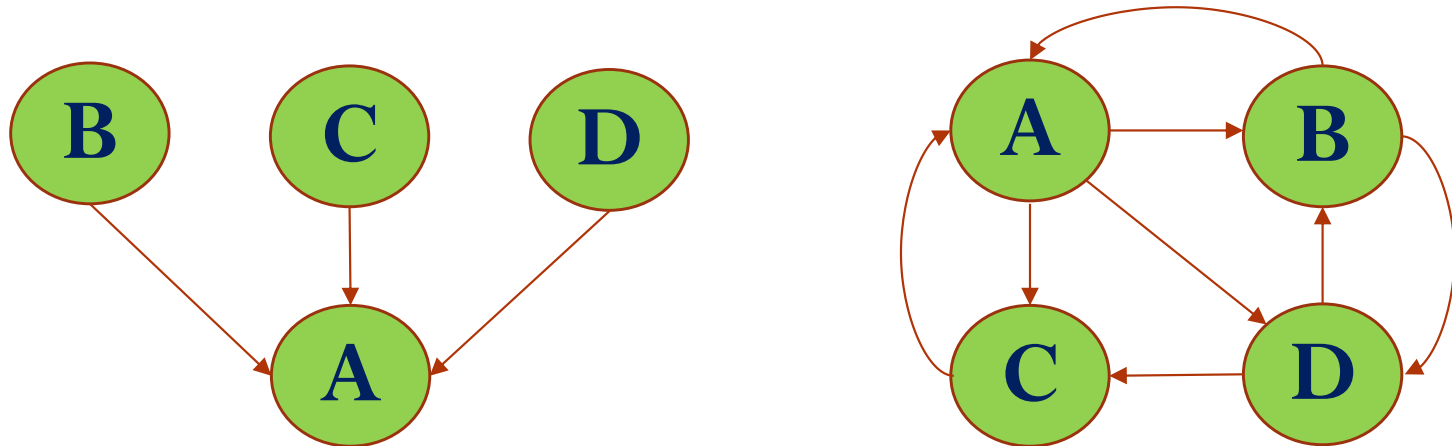
- The PageRank (PR) of page i is the **probability** that a surfer following the **random walk** has arrived on i at some distant point in the future

PageRank on directed graphs

- The PageRank (PR) of page i is the **probability** that a surfer following the **random walk** has arrived on i at some distant point in the future
- **Motivation:**
 - Users of Web “vote with their feet”
 - Behavior of a random surfer indicates which pages users likely to visit

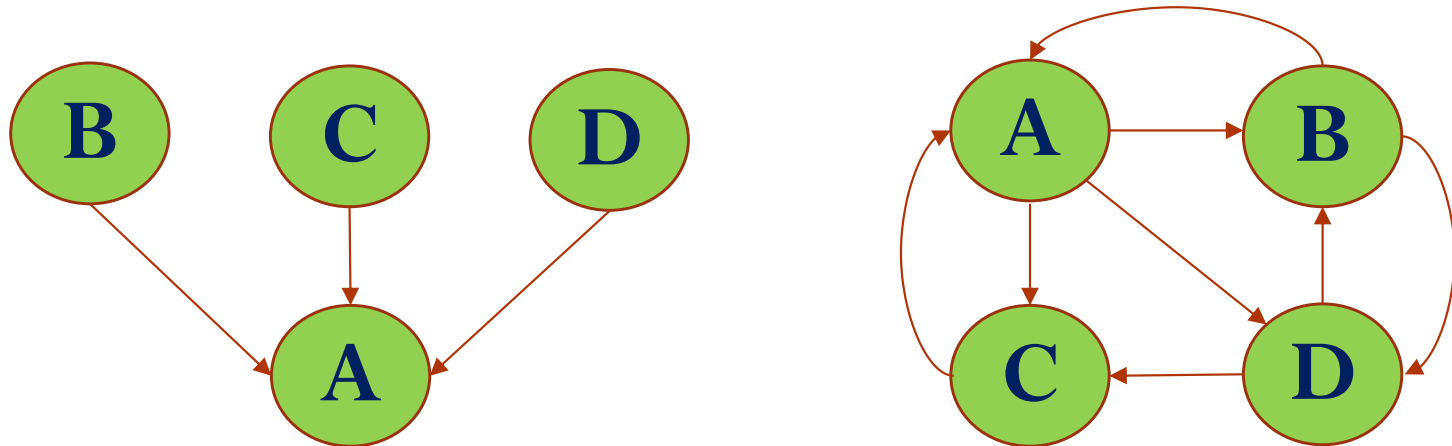
PageRank on directed graphs

- Compute PageRank by simulating random surfers



PageRank on directed graphs

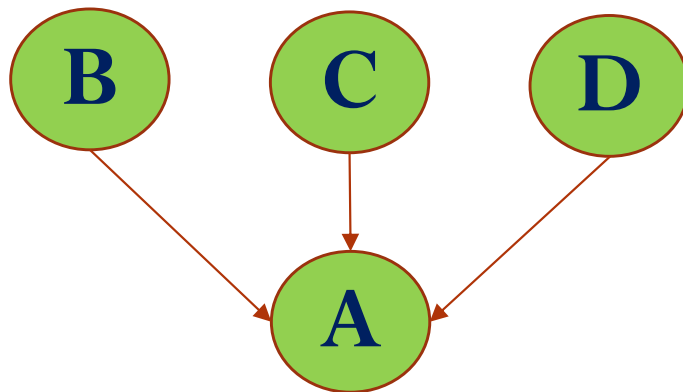
- Compute PageRank by simulating random surfers



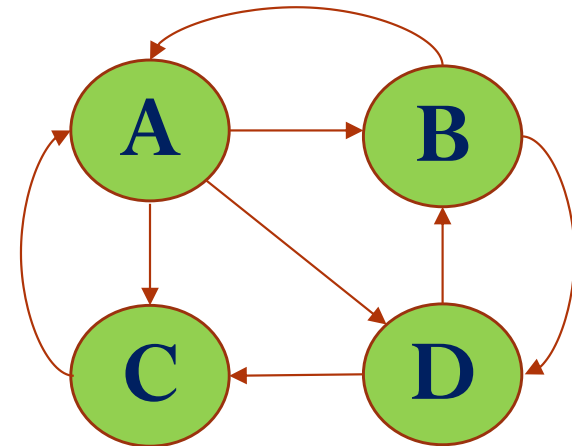
$$\text{Pr}[A] = \text{Pr}[B] + \text{Pr}[C] + \text{Pr}[D]$$

PageRank on directed graphs

- Compute PageRank by simulating random surfers



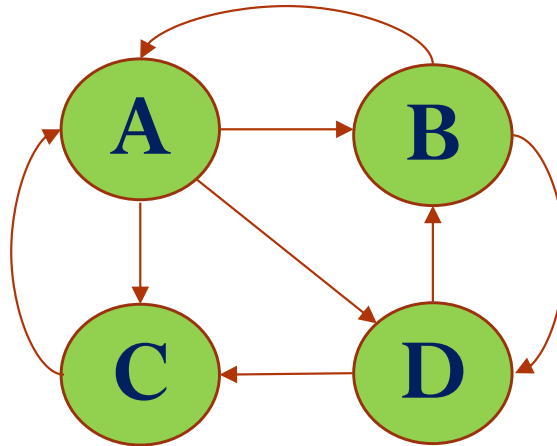
$$\Pr[A] = \Pr[B] + \Pr[C] + \Pr[D]$$



Complicated ☹️

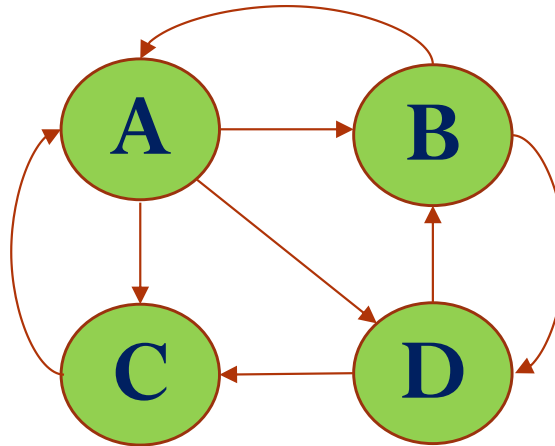
PageRank on directed graphs

- Compute PageRank by simulating random surfers



PageRank on directed graphs

- Compute PageRank by simulating random surfers



- If a random surfer starts from A, it will next be at B/C/D with probability $1/3$
- A random surfer at B will next be at A/D with probability $1/2$
- ...

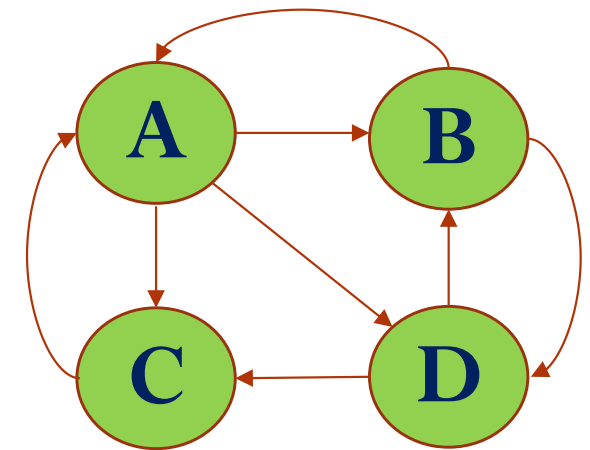
PageRank on directed graphs

- Transition matrix:

$$M = \begin{matrix} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix} \end{matrix}$$

Probability a
surfer goes
from A to D

A



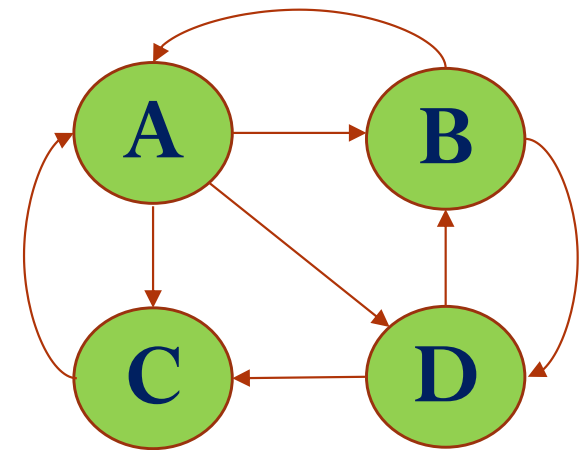
PageRank on directed graphs

- Transition matrix:

$$M = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}$$

Probability a
surfer goes
from A to D

A



- Initialize a starting point of surfer as a vector $\mathbf{v}_0$
- $M\mathbf{v}_i$ is the distribution of the surfer at the step $i+1$. Repeat until convergence.

PageRank on directed graphs

$$\begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} 9/24 \\ 5/24 \\ 5/24 \\ 5/24 \end{bmatrix} \begin{matrix} \leftarrow A \\ \leftarrow B \\ \leftarrow C \\ \leftarrow D \end{matrix}$$

M
 v_0
 v_1

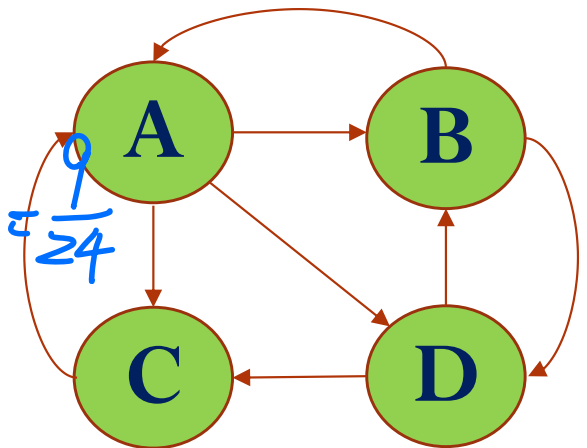
$$\frac{1}{8} + \frac{1}{4} = \frac{12}{32} = \frac{6}{16} = \frac{3}{8} = \frac{9}{24}$$

$$A \rightarrow A \quad \frac{1}{4} \times 0$$

$$B \rightarrow A \quad \frac{1}{4} \times \frac{1}{2}$$

$$C \rightarrow A \quad \frac{1}{4} \times 1$$

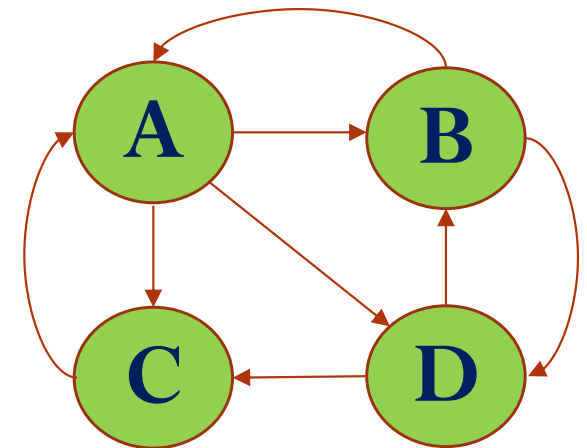
$$D \rightarrow A \quad \frac{1}{4} \times 0.$$



PageRank on directed graphs

$$\begin{array}{c} \text{A} \rightarrow \end{array} \begin{bmatrix} \boxed{0} & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} \boxed{9/24} \\ 5/24 \\ 5/24 \\ 5/24 \end{bmatrix} \begin{array}{c} \leftarrow \text{A} \\ \leftarrow \text{B} \\ \leftarrow \text{C} \\ \leftarrow \text{D} \end{array}$$

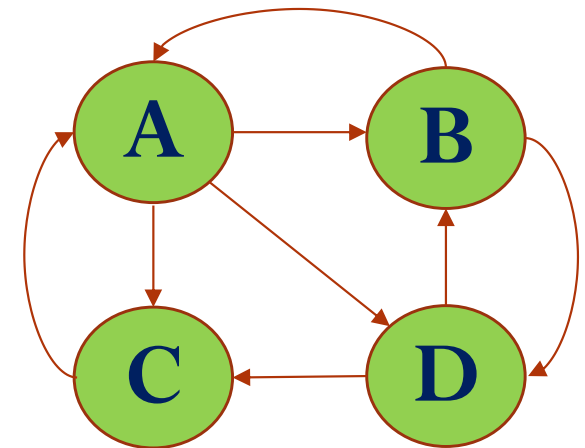
$\mathbf{M}$
 $\mathbf{v}_0$
 $\mathbf{v}_1$



PageRank on directed graphs

$$\begin{array}{c} \text{A} \rightarrow \end{array} \underbrace{\begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}}_{\mathbf{M}} * \underbrace{\begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}}_{\mathbf{v}_0} = \underbrace{\begin{bmatrix} 9/24 \\ 5/24 \\ 5/24 \\ 5/24 \end{bmatrix}}_{\mathbf{v}_1} \begin{array}{c} \leftarrow \text{A} \\ \leftarrow \text{B} \\ \leftarrow \text{C} \\ \leftarrow \text{D} \end{array}$$

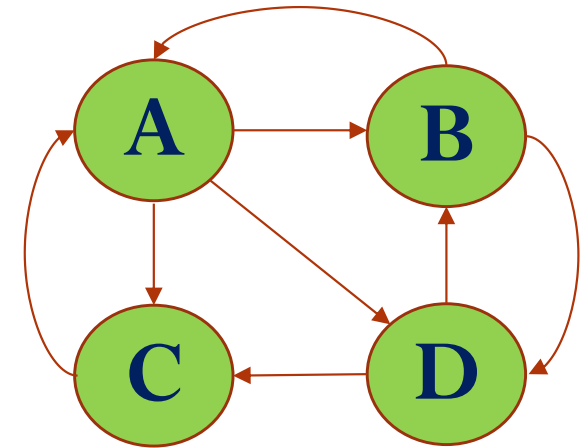
- Surfer at A goes to A with prob. $\frac{1}{4} * 0$
- Surfer at B goes to A with prob. $\frac{1}{4} * \frac{1}{2}$
- Surfer at C goes to A with prob. $\frac{1}{4} * 1$
- Surfer at D goes to A with prob. $\frac{1}{4} * 0$
- Surfer will be at A with prob. $\frac{1}{4} * \frac{1}{2} + \frac{1}{4} * 1 = 9/24$



PageRank on directed graphs

$$\begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} 9/24 \\ 5/24 \\ 5/24 \\ 5/24 \end{bmatrix}$$

$\mathbf{M}$ $\mathbf{v}_0$ $\mathbf{v}_1$



- Final result:

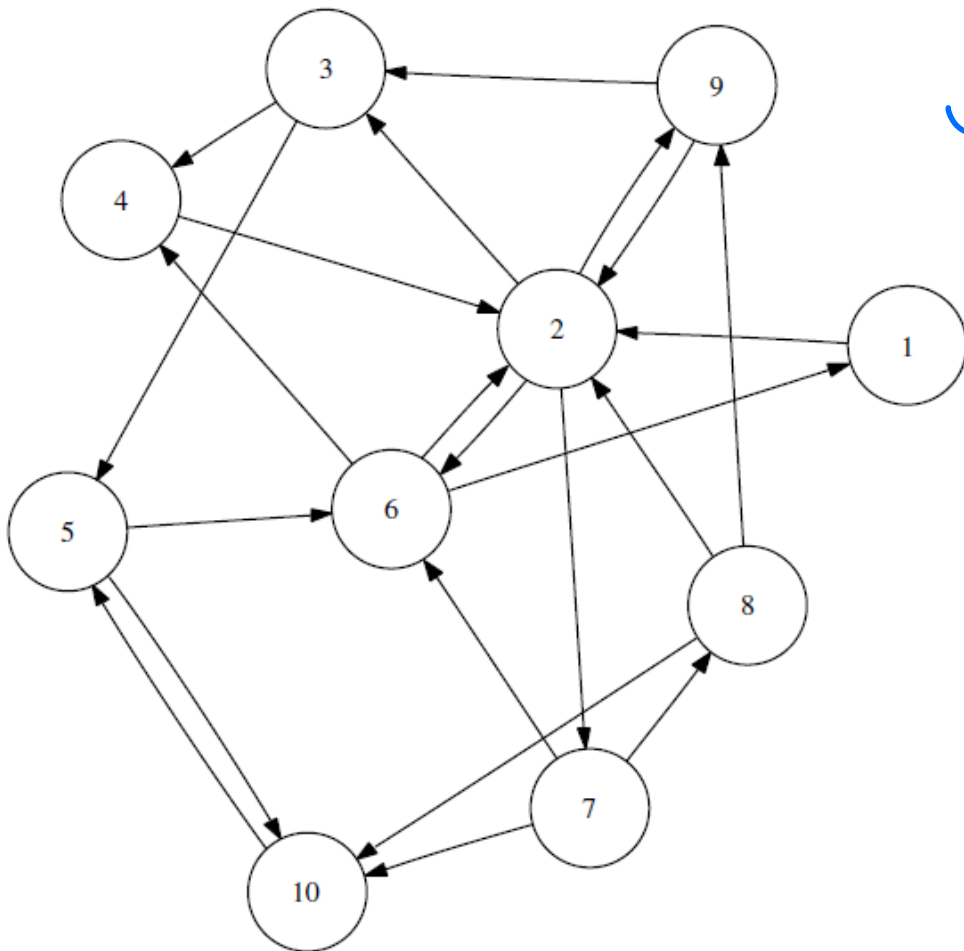
$$\begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}, \begin{bmatrix} 9/24 \\ 5/24 \\ 5/24 \\ 5/24 \end{bmatrix}, \begin{bmatrix} 15/48 \\ 11/48 \\ 11/48 \\ 11/48 \end{bmatrix}, \begin{bmatrix} 11/32 \\ 7/32 \\ 7/32 \\ 7/32 \end{bmatrix}, \dots, \begin{bmatrix} 3/9 \\ 2/9 \\ 2/9 \\ 2/9 \end{bmatrix}$$

$\mathbf{v}_0$ $\mathbf{v}_1$ $\mathbf{v}_2$ $\mathbf{v}_3$ $\mathbf{v}_n$

0次点击

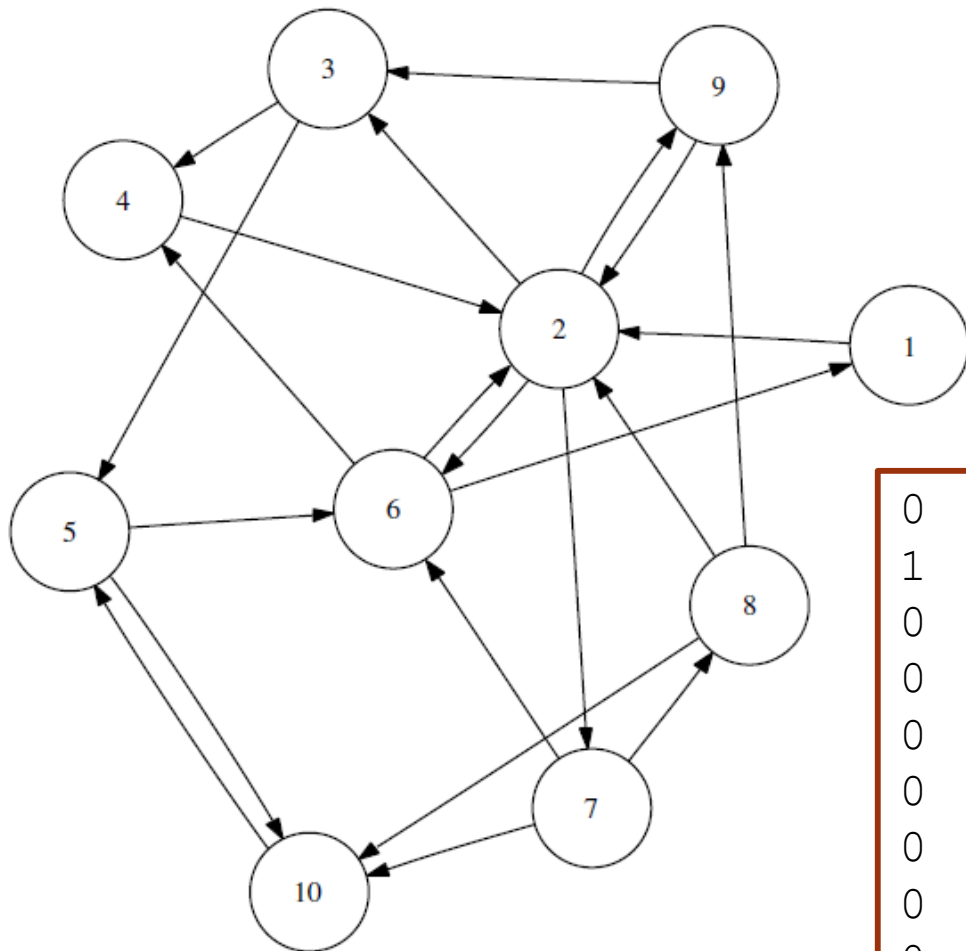
1次点击

Exercise: Form the transition matrix



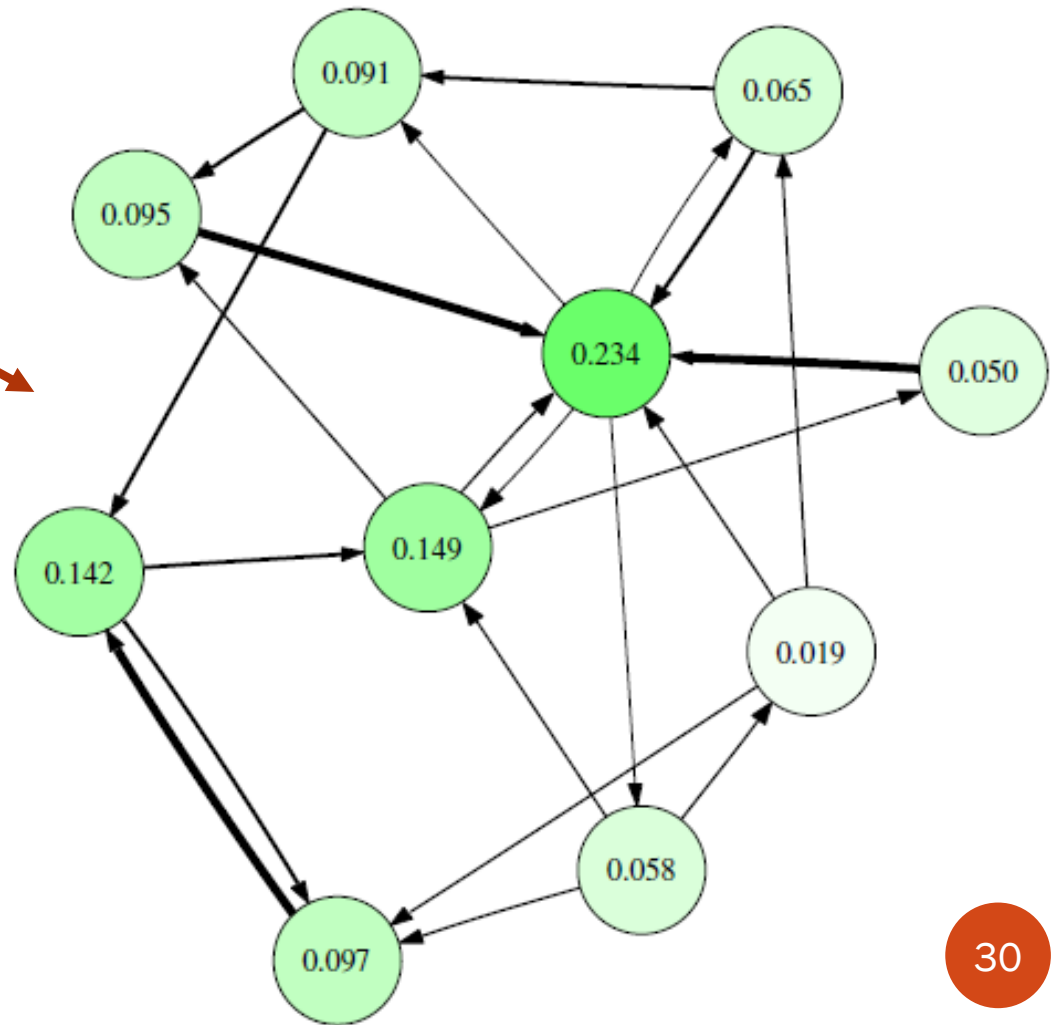
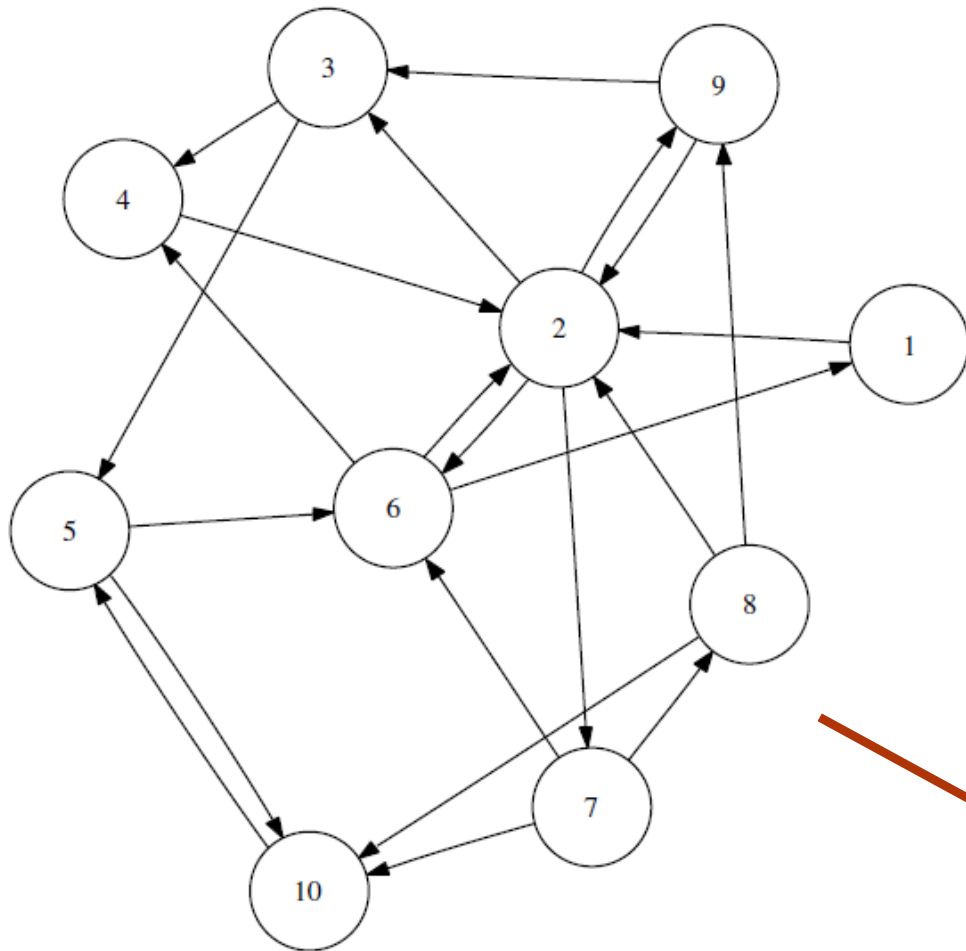
10 x 10.

Exercise: Form the transition matrix



0	0	0	0	0	1/3	0	0	0	0
1	0	0	1	0	1/3	0	1/3	1/2	0
0	1/4	0	0	0	0	0	0	1/2	0
0	0	1/2	0	0	1/3	0	0	0	0
0	0	1/2	0	0	0	0	0	0	1
0	1/4	0	0	1/2	0	1/3	0	0	0
0	1/4	0	0	0	0	0	0	0	0
0	0	0	0	0	0	1/3	0	0	0
0	1/4	0	0	0	0	0	1/3	0	0
0	0	0	0	1/2	0	1/3	1/3	0	0

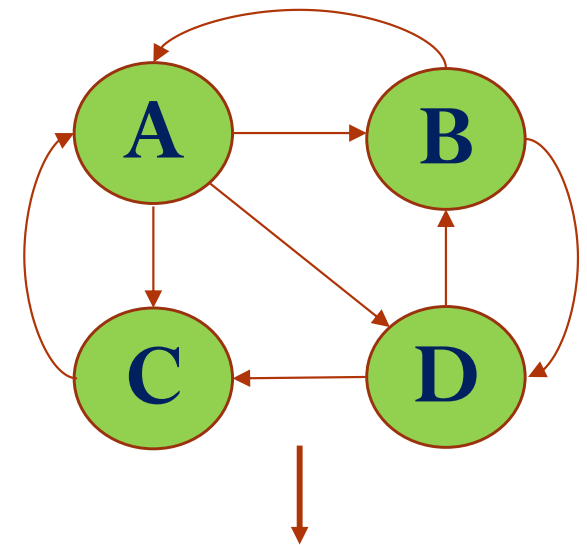
Exercise: Implement PageRank (Python)



Convergence

- Why does PageRank converge independently from any $\mathbf{v}_0$?

$$M = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}$$

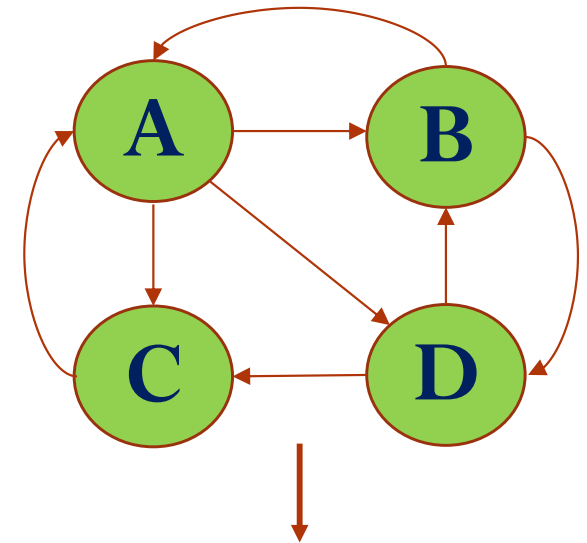


$$\begin{bmatrix} 3/9 \\ 2/9 \\ 2/9 \\ 2/9 \end{bmatrix}$$

Convergence

- Why does PageRank converge independently from any $\mathbf{v}_0$?

$$M = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}$$



- If $\mathbf{M}$ is a stochastic, irreducible and aperiodic
 - PageRank will converge to the eigenvector corresponding to the largest eigenvalue 1

特征向量

$$\begin{bmatrix} 3/9 \\ 2/9 \\ 2/9 \\ 2/9 \end{bmatrix}$$

PageRank with damp

- PR ^{收敛} does not always converge, or convergence may not be unique in real-world web graphs
- To fix this, the random surfer can at each step randomly jump to any page of the Web with some probability $1 - d$

$$\text{Pr}[i] = \lim_{k \rightarrow \infty} (dM + (1 - d)U)^k v)_i$$

where d is called the **damping factor**, U is the matrix with all $1/n$ values where n is the number of vertices, v is the initialization of random surfer.

Using PR to score query results

- PageRank: **global** score, independent of the query
- Can be used to raise the weight of **important** pages:

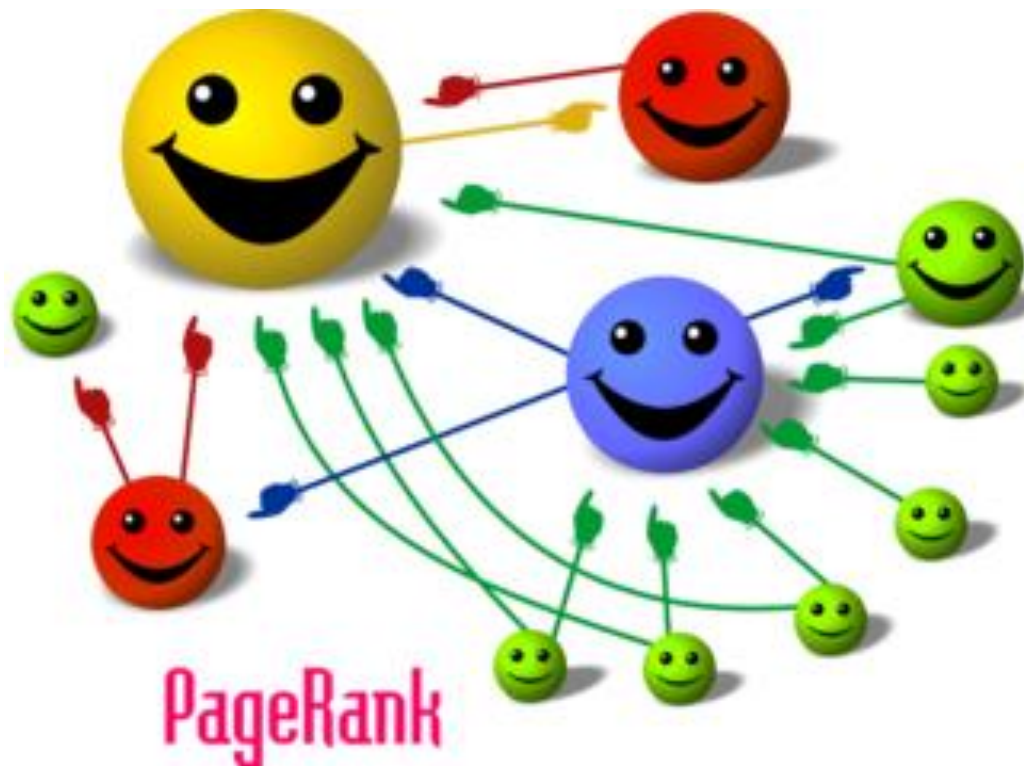
$$\text{weight}(\mathbf{t}, \mathbf{d}) = \text{tf-idf}(\mathbf{t}, \mathbf{d}) \times \text{pr}(\mathbf{d}),$$

where $\text{tf-idf}(\mathbf{t}, \mathbf{d})$ present the weight of the query term $\mathbf{t}$ on the webpage $\mathbf{d}$ and $\text{pr}(\mathbf{d})$ is the page rank of $\mathbf{d}$

- This can be directly incorporated **in the inverted index**

Homework

- Write Python script to compute PageRank of the graph



<https://en.wikipedia.org/wiki/PageRank>